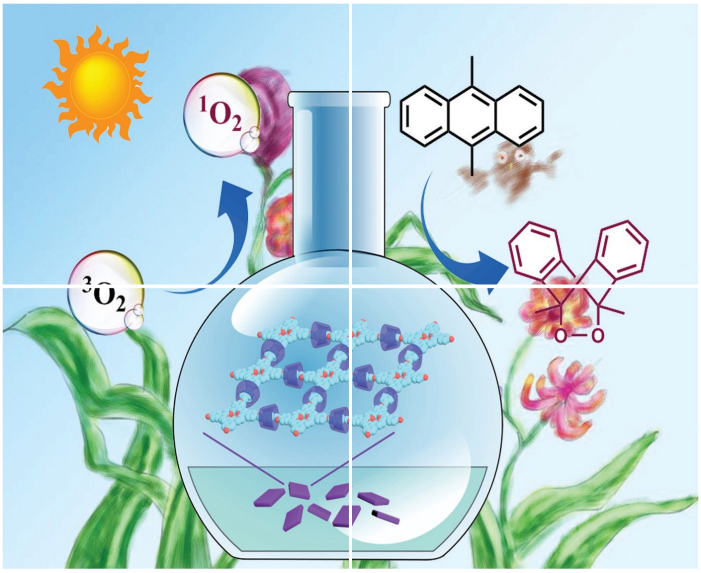
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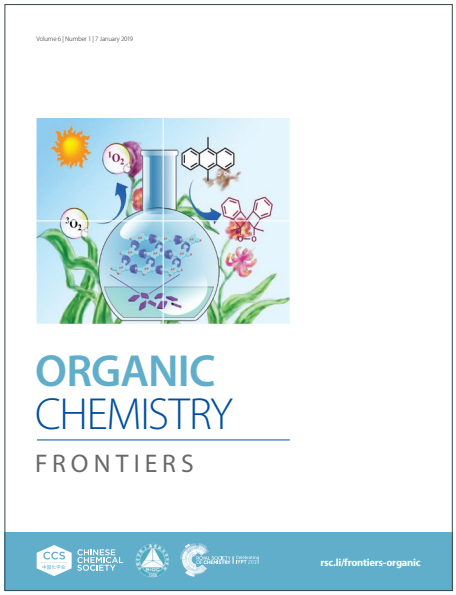
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ARTICLE

NHC-Catalyzed β-Specific Addition of N-Based Nucleophiles to Allenoates

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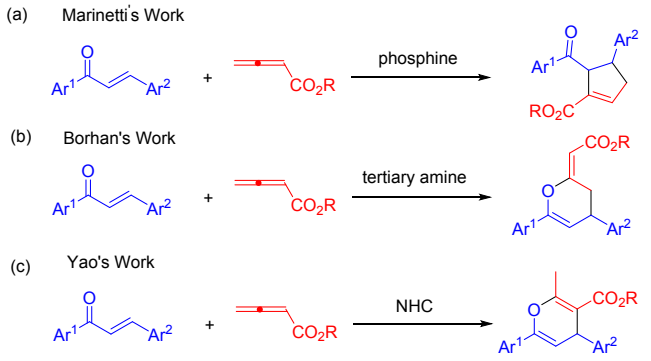
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N-heterocyclic carbene (NHC) catalyzed regiospecific β-additions of nitrogen-containing nucleophiles including nitrogenous heterocycles and trifluoromethylated acylhydrazone to allenoates were accomplished under mild reaction conditions, harnessing NHC as a Lewis base to furnish branched N-alkyl compounds in moderate to good yields with high E/Z selectivity and broad substrate scope. DFT calculations on the possible reaction mechanisms show that β-addition pathway is more energetically favourable than γ-addition pathway, which is in agreement with the experimental observations.

Introduction

Allenes have exhibited high reactivity in comparison to the similar alkenes due to their cumulated C-C double bonds, thus they have been demonstrated as useful building blocks for the construction of complex skeletons.[1] After the pioneering work of Lu and co-workers,[2] allenoates have been proven to undergo many types of annulations including [3+2] and [2+n] cyclization to establish carbo- or heterocyclic scaffolds.[3] In particular, the transformations of allenoates under the catalysis of nucleophilic amine[4] and phosphine[5] have received considerable interest due to the flexibility of modifying molecular complexity.[6] In 2008, Marinetti's group reported a phosphine-catalyzed highly enantioselective [3+2] cyclizations of ethyl 2,3-butadienoate with α,β-unsaturated esters or ketones (Scheme 1a).[7] Furthermore, the same reaction promoted by a tertiary amine gave pyran derivatives through β,γ-[2+4] annulation disclosed by Borhan's group at a later time (Scheme 1b).[8] So far, N-heterocyclic carbenes (NHCs) have been extensively utilized as nucleophilic catalysts to facilitate the conversions of aldehydes,[9] acids,[10] esters,[11] α,β-unsaturated aldehydes,[12] etc.[13] Very recently, the above-mentioned reactions catalyzed by NHC performed by our group gave the α,β-[2+4] cycloadduct with regioselectivity distinct from the annulations facilitated by phosphine or tertiary amine (Scheme 1c).[14] Therefore, we can conclude safely that the structure of the products and the reaction selectivity of allenoate-involved transformations are mainly reliant on the catalysts.

Scheme 1 Annulation of allenoates with chalcones catalyzed by nucleophiles.



Recently, the N-alkylation of nitrogen-containing heterocycles catalyzed by nucleophilic organic molecular catalysts has received enormous attention.[15] Many nitrogenous heterocycles gracefully exhibit superior biological characteristics, such as Losartan,[16] Etomidate[17] and Ruxolitinib.[18] And some pyrazole-based Janus kinase (JAK) inhibitors were developed successfully (Fig. 1).[19] In 2014, Breit and coworkers reported the first atom-economic rhodium-catalyzed regioselective α-umpolung coupling of allenes with imidazoles (Scheme 2a).[20] Later, a phosphine-catalyzed γ-addition reaction of allenoates with pyrazole and imidazole derivatives through umpolung process was disclosed by Virieux's and Guo's group (Scheme 2b).[21] To continue our work on NHC-catalyzed transformations of allenoates and study on the dependence of their regioselectivity on catalyst further,[14,22] we carried out NHC-promoted reactions of nitrogenous heterocycles with allenoates, which provided regiospecific β-adducts under mild conditions with excellent atom economy (Scheme 2c). These results also paved a new

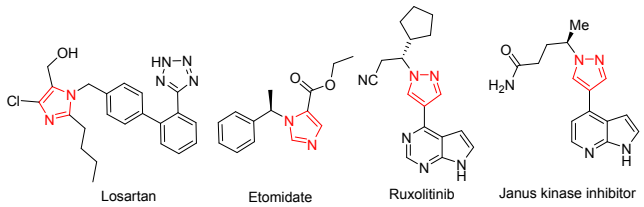
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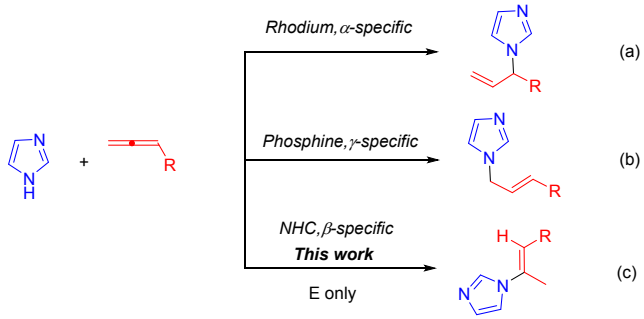
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avenue to the derivatization of N-containing heterocycles with biological significance.

Figure 1 Examples of bioactive imidazole and pyrazole derivatives.



Scheme 2 Reactions of imidazole with allene presented by catalysts.



Results and discussion

We commenced our study with the model reaction of imidazole (**1a**) with 2,3-butadienoic acid, ethyl ester (**2a**), and the results were summarized in Table 1. In the presence of precatalysts **A** (20 mol%) and Cs_2CO_3 (50 mol%), the β-addition reaction occurred smoothly and provided the corresponding product **3aa** in 54% yield as a single geometrical isomer in THF (Table 1, entry 1). The single crystal X-ray diffraction confirmed the structure of **3aa** unambiguously (Fig. 2).[23] Other NHC precursors, including imidazolium salts **B-D**, triazolium salts **F-H** and thiazolium catalyst **E**, resulted in low yield or produced a mixture of E and Z isomers (entry 2-8). After a careful screening of the solvents and bases, compound **3aa** was obtained regiospecifically in 58% yield in the presence of ^{t}BuOK in THF (entry 17). Besides, a blank test was carried out and the reaction could not occur without NHC catalyst (entry 18). Gratifyingly, further assessment on reaction temperature revealed the reaction yield was slightly improved by elevating the temperature to 45 °C, which was finally established as the optimal reaction conditions (entry 20).

Figure 2 X-ray structure of product **3aa**.

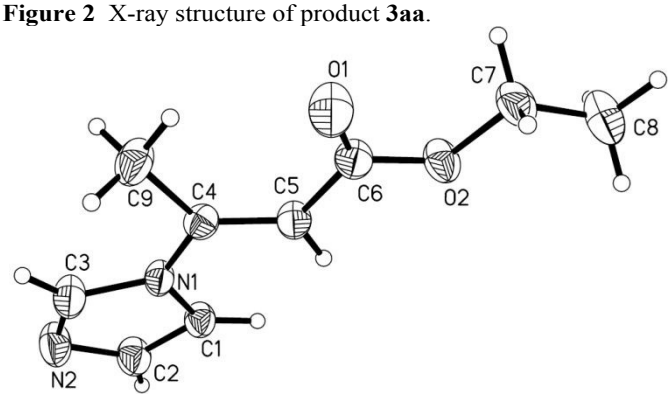
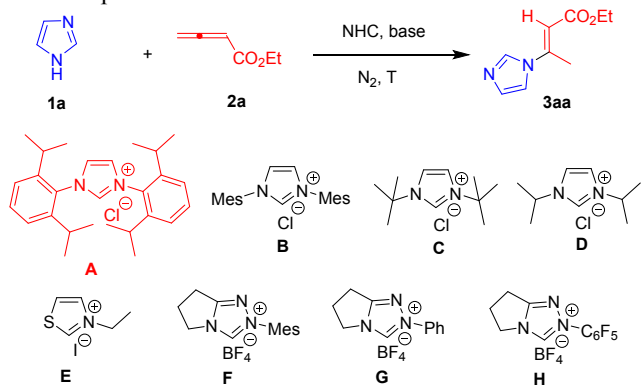


Table 1 Optimization of the Reaction Conditions[a]



Entry	NHC	Base	Solvent	Yield (%)[b]	E/Z[c]
1	**A**	Cs_2CO_3	THF	54	E only
2	**B**	Cs_2CO_3	THF	43	E only
3	**C**	Cs_2CO_3	THF	50	E only
4	**D**	Cs_2CO_3	THF	21	E only
5	**E**	Cs_2CO_3	THF	trace	ND
6	**F**	Cs_2CO_3	THF	36	15:1
7	**G**	Cs_2CO_3	THF	38	> 20:1
8	**H**	Cs_2CO_3	THF	25	> 20:1
9	**A**	Cs_2CO_3	DCM	42	> 20:1
10	**A**	Cs_2CO_3	DCE	31	> 20:1
11	**A**	Cs_2CO_3	DMSO	39	E only
12	**A**	Cs_2CO_3	MeCN	44	E only
13	**A**	Cs_2CO_3	$CHCl_3$	45	E only
14	**A**	DMF	THF	trace	ND
15	**A**	K_2CO_3	THF	29	> 20:1
16	**A**	DABCO	THF	trace	ND
17	**A**	^{t}BuOK	THF	58	E only
18		^{t}BuOK	THF	NR	ND
19[d]	**A**	^{t}BuOK	THF	65	E only
20[e]	**A**	^{t}BuOK	THF	76	E only
21[f]	**A**	^{t}BuOK	THF	60	E only

[a]Reaction conditions: **1a** (0.20 mmol), **2a** (0.30 mmol), NHC (20 mol%), base (0.10 mmol), solvent (2.0 mL) at 25 °C. [b]Isolated yields. [c]Determined by ^{1}H NMR. [d]Reaction performed at 35 °C. [e]Reaction carried out 45 °C. [f]Reaction conducted at 55 °C. ND = not determined. NR = no reaction.

With the optimized conditions in hand, the substrate scope of the reaction was evaluated (Table 2). Initially, it was found that a broad range of nitrogen-containing heterocycles **1** were compatible with the reaction conditions such as imidazole, pyrazole, triazole and aminotetrazole. We were pleased to find

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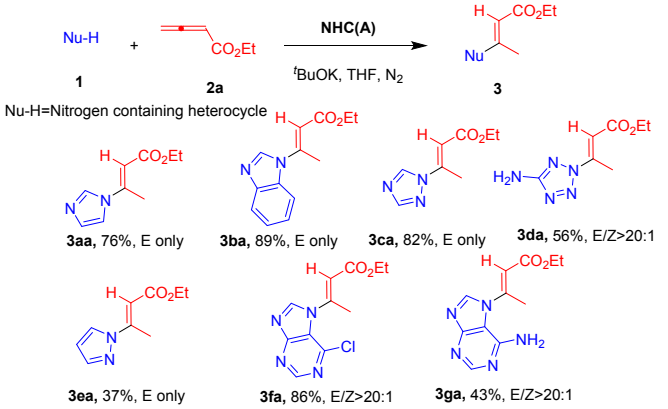
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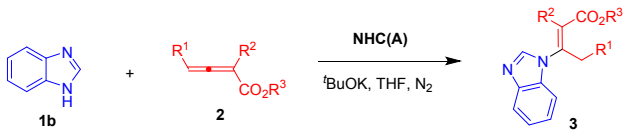
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that adenine and guanine was also tolerated well and all of them gave the corresponding products with moderate to good yields and stereoselectivity (Table 2, **3aa-3ga**). Unfortunately, when pyrrolidine, piperidine, pyrrole, indole, phthalimide and isatin were subject to this protocol, no conversion of the starting materials was discovered. Thus, we currently concluded safely that the nitrogen-containing nucleophiles with pKa ranging from 2.27 to 7.47 could take part in the process and deliver the β-specific adduct readily. Next, the scope of allenoates prepared readily from either commercial or known starting materials was explored. These reactions worked efficiently to give the analogous products in good yields with exclusive E-selectivity (Table 2, **3bb-3bh**). However, we found that the allenoates possessing smaller ester substituents such as Me and Et, gave the products in higher yields (**3bb** vs **3bc** & **3bd** in Table 2). In addition, the β-specific addition reaction between γ-substituted allenoate and nitrogen-containing nucleophile went readily and gave the product in 87% yield (Table 2, **3bh**).

Table 2 Exploration of Substrate Scope[a]



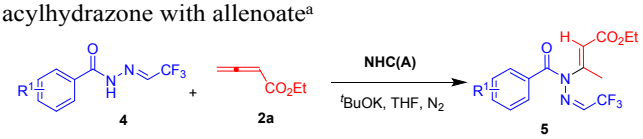


3bb, $R^1/R^2/R^3$=H//H/Me, 94%, E only
3bc, $R^1/R^2/R^3$=H//H/Bn, 73%, E only
3bd, $R^1/R^2/R^3$=H//H/^{n}Bu, 62%, E only
3be, $R^1/R^2/R^3$=H/Bn/Et, 56%, E only
3bf, $R^1/R^2/R^3$=H//H/Cy, 64%, E only
3bg, $R^1/R^2/R^3$=H//H/^{s}Bu, 58%, E only
3bh, $R^1/R^2/R^3$=Bn//H/Et, 87%, E only

[a]Reaction conditions unless otherwise specified: **1** (0.20 mmol), **2** (0.30 mmol), NHC (20 mol%), base (0.10 mmol), dry THF (2.0 mL) at 45 °C.

Trifluoromethylated molecules have become a research focus in the fields of pesticides and medicines due to their notable properties, thus much effort has been devoted to the conversion of building blocks with trifuloromethyl group to the new chemical entities with specific functions or properties.[24] Meanwhile, N-acylhydrazone (NAH) is a privileged and ubiquitous core distributed broadly in hit and lead compounds acting on numerous types of molecular targets.[25] In 2013, Huang's group reported the first phosphine-catalyzed tunable aza-Michael β-addition reaction of allenoates with trifluoromethylated acylhydrazones and obtained two isomerized products.[26] To further expand the substrate scope, the trifluoromethylated acylhydrazone was also tested for the reaction (Table 3). To our delight, these trifluoromethylated substrates were compatible with the reaction conditions and provided the single products in good yields with high E/Z selectivities as well (Table 3, **5aa-5ga**). It should be also noteworthy that electron-donating groups on the phenyl ring of acylhydrazone such as OMe and Me reduced the product yields of **5ca** (33%) and **5fa** (45%) slightly, respectively.

Table 3 NHC-catalyzed reaction of trifluoromethylated acylhydrazone with allenoate[a]

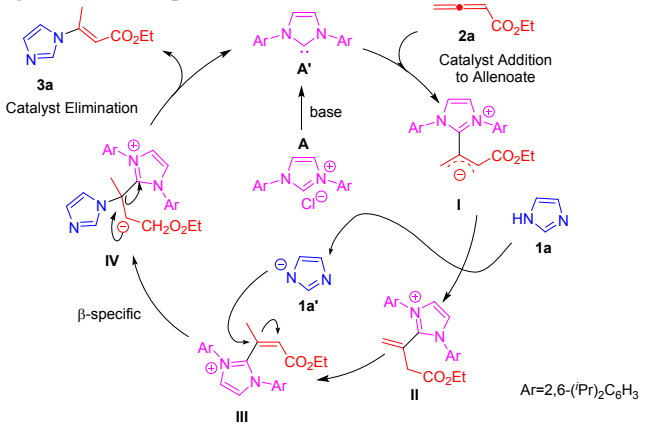


5aa, R^1=H, 68%, E only
5ba, R^1=4-Br, 64%, E only
5ca, R^1=4-MeO, 33%, E/Z>20:1
5da, R^1=4-F, 72%, E/Z>20:1
5ea, R^1=4-^{t}Bu, 74%, E only
5fa, R^1=2-Me, 45%, E/Z>20:1
5ga, R^1=3-Cl, 52%, E only

[a]Reaction condition unless otherwise specified: **4** (0.20 mmol), **2** (0.30 mmol), NHC (20 mol%), base (0.10 mmol), dry THF (2.0 mL) at 45 °C.

Product **3aa** was employed to illustrate the reaction process, and a plausible catalytic cycle for this reaction was depicted in Figure 3. Firstly, the deprotonation of precatlyst **A** generated NHC **A'**, which then added to the allenoate **2a** to form intermediate **I**. Then **I** reacted with **1a** to give **1a'** and intermediate **II**, which isomerized into **III** with a conjugate system subsequently. Nitrogen-containing nucleophiles and intermediate **III** underwent a β-specific addition to give intermediate **IV**, which was then transformed into the desired product **3aa** through elimination and the release of catalyst.

Figure 3 Our Proposed Mechanism



On the basis of our observation, some control experiment were carried out to shed some light on the reaction mechanism. Firstly, NHC was mixed with the same equivalent **1a** in $CDCl_3$, and no decrease of the amount of **1a** was detected from the spectrum of ^{1}H NMR. Then, the amount of **2a** was reduced by 54.5% after the introduction of same equivalent NHC into its solution indicated by NMR data. The formation of the final product **3aa** was determined by adding **1a** into the abovementioned reaction system in 15% yield (see Supporting Information). Thus there should be an intermediate generated from NHC and **2a** initially, which reacted with **1a** to give product **3aa** eventually. These results provided some

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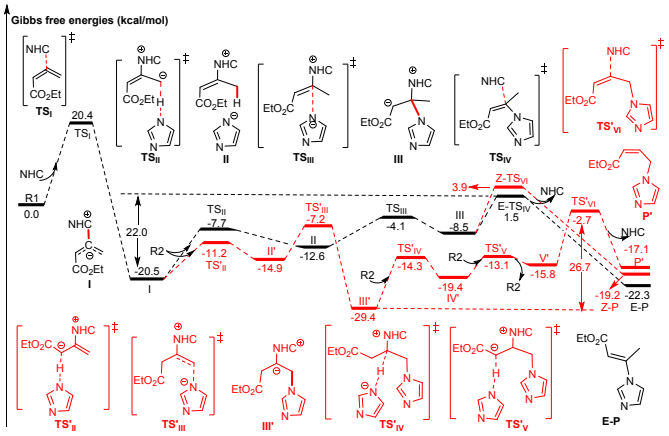
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experimental support for our proposed plausible reaction mechanism.

To get some insight into the reaction pathway and chemoselectivity, we have performed DFT calculations by using Gaussian16 program, and more computational details can be found in Supporting Information. As shown in Figure 4, there are two competing reaction pathways associated with the structural transformations from **R1** and **R2** to **E/Z-P** and **P'** through two different pathways, respectively. The calculated results show that NHC can nucleophilically attack on **R1** via transition state $\mathbf{TS_I}$ with an energy barrier of 20.4 kcal/mol to form intermediate **I**, then the different pathways divorced in the following steps. One pathway undergoes γ-C protonation by **R2** via transition state $\mathbf{TS_{II}}$, β-C-N bond formation via transition state $\mathbf{TS_{III}}$, dissociation of catalyst and product **E-P** via transition state $\mathbf{E\text{-}TS_{IV}}$. It should be noted that the Z-isomer pathway associated with $\mathbf{E\text{-}TS_{IV}}$ and **Z-P** is obviously unfavourable in the energy profile, which is consistent with the experimental observation. The other pathway occurs through α-C protonation by **R2** via transition state $\mathbf{TS'_{II}}$, γ-C-N bond formation via transition state $\mathbf{TS'_{III}}$, **R2**-assisted [1,2]-proton transfer via transition states $\mathbf{TS'_{IV}}$ and $\mathbf{TS'_{V}}$, and dissociation of catalyst and product **P'** via transition state $\mathbf{TS'_{VI}}$. The energy barrier of entire reaction associated with E-**P** is 22.0 kcal/mol, which is lower than that associated with **P'** (26.7 kcal/mol), indicating the **E-P** should be the main product. All the calculations are in agreement with the experimental results.

Figure 4 Energy profiles on the competing reaction pathways (All the energies were computed at the M06-2X/6-31G**/IEF-PCM_{THF} level of theory)



Conclusions

In conclusion, we have described a NHC-catalyzed β-specific addition of nitrogen-based pronucleophiles to allenoates, offering a new approach towards the heterocyclic and fluorine-containing substituted Michael olefins with excellent atom economy. This protocol represented a new reaction pattern of NHC-catalyzed transformations involving allenoates. DFT calculations on the possible reaction pathways show that the formation of β-adduct is more energetically favourable than that of γ-adduct, which is in accordance with the results of the experiment. Further studies on the application of this methodology are underway in our lab.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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